CS 598 (Fall 2026)
Project Proposal

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Problem Statement:
Consider human $\mathcal{H}$ who needs to make a move $k \in [K]$ from a state $x \in \mathbb{R}^n$.
The ground truth utility matrix: $U = \begin{pmatrix} -u_1- \\ \vdots \\ -u_K- \end{pmatrix} \in \mathbb{R}^{K \times n}$, gives us the utility for each move based on state: $u(x) = \begin{bmatrix} u_1 x \\ \vdots \\ u_K x \end{bmatrix} = Ux$. The optimal move from state x is thus: $y^* = \arg\max_i(u_i x)$.
At time t, the human has a mental weight matrix of the utilities $H_t \in \mathbb{R}^{K \times n}$, which it uses to make a move: $\hat{y} = \arg\max_i(Hx)_i$.

The human and algorithm $\mathcal{A}$ interact as follows at timestep $t \geq 0$:

1. $\mathcal{A}$ picks an action subset $A_t \subseteq [K]$ with $c'_A \leq |A_t| \leq c_A$ and a feature subset $F_t \subseteq [n]$ with $|F_t| \leq c_F$ to show to $\mathcal{H}$
2. $\mathcal{H}$ observes $(x_t)_i$ for $i \in F_t$ and A_t and makes a prediction $\hat{y} = \arg\max_{k \in A_t} \hat{u_t}(k)$, where human's estimated utility for move k is $\hat{u_t}(k) = \sum_{j \in F_t} (H_t)_{k,j}(x_t)_j$ and exhibits a loss according to the Decision Regret loss: $\ell_{\text{regret}}(x_t, \hat{y_t}) = \max_k (Ux_t)_k - (Ux_t)_{\hat{y_t}}$
3. $\mathcal{H}$ updates H_{t+1} for all observed features F_t and actions A_t, based on some arbitrarily learning rule which where $(H_t)_{k,j}$ converges to the true $(U)_{k,j}$ as the number of times feature j and move k is observed tends to infinity

Here, $\mathcal{A}$ must select subsets $F = (F_t)_{t \geq 0}$ and $A = (A_t)_{t \geq 0}$, minimizing discounted loss of human's predictions over time: $\sum_{t=0}^{\infty} \delta^t \ell_{\text{regret}}(x_t, \hat{y_t})$, where $\delta \in (0, 1)$.

Questions:
What is the optimal sequence of subsets of actions and features that $\mathcal{A}$ should suggest to minimize this loss? Are there certain matrices U which result in better conditions of convergence/learning? What happens when features x_i are correlated, and when actions are correlated?

Background Literature:
Our work is based on the research from Noti et al[2] where AI-assistance for human learning is discussed. They study the circumstance where human makings the final decision in a human-AI decision process. In their model, ground truth is defined as a linear function $y = c + \sum_{i=1}^{n} a_i x_i$ over n independent, standardized features. For each round of decision making, the algorithm could choose up to k feature subset to display to the human. The human would then use their own possibly incorrect coefficient vector h_t to make the prediction. However, in choosing the k features, there exist a trade off between informativeness and divergence, or how much information does it reveal about the ground truth, and how accurately could humans understand it. Noti et al expressed the loss through the Mean Squared Error between human prediction and ground truth.

Noti et al. made the problem tractable with the assumption that through repetitive interaction with the AI, human's belief would gradually converge to the ground truth by a monotone function ϕ. They demonstrate a second tradeoff, of the tension between achieving the best result under the "fixed mindset" and sacrificing the short term accuracy for long learning of "growth mindset". Their work produced a surprisingly simple solution to minimizing the loss: since the features are additive, the optimal solution is selecting the same subset at every round.

However, Guan et al. [1] pointed out how this fails when the features are no longer independent, and that once there exists correlation between features, displaying any feature might leak information about the other features and since humans have never observed the correlated feature directly this might leave persistent error that no fixed subset could fix. Therefore a stationary policy could be arbitrarily worse than optimal. Under this setting, the solution would be NP-hard rather than $O(n \log n)$ in the previous work.

Importance and Novelty

To build intuition for our setting, consider the example of Noti et al. [2]: a doctor orders a limited number of tests and then predicts a risk score. In practice, however, diagnosis is only an intermediate step, and the final step is for the doctor to *take action*. Now lets consider a patient with a runny nose. As in Noti et al., the algorithm chooses which tests the doctor sees (e.g. anything from congestion severity to a neurological exam). In our setting, the algorithm also chooses which treatments are on the table (e.g. daytime cold medicine, night-time cold medicine, or surgery!). The doctor's job is now not to estimate a risk score but to pick the best treatment, and therefore each row of the utility matrix U describes how one treatment's value depends on the test results.

More generally, we look at extending work done by [1, 2] with a more natural game-like formulation of this decision process, in which an AI is assisting the human. Instead of predicting for one value, the human is trying to pick the best move to make, given a specific state, where the optimal move is dependent on a utility matrix which is unknown and to be learnt. Thus this poses a more relevant and interesting question, which has not been explored yet.

Feasibility

Under this general framework of AI-assisted decision making for the best move, we have listed a number of questions we wish to investigate, above. We aim to attempt this through first looking at this problem with only one possible move ($K = 1$) and testing whether we can use the theory from [1, 2] to solve this case. Then we will work in the case where features are all drawn independently from a normal distribution, and work on the problem with a 2-dimensional action space ($K = 2$), after which we should be able to generalize to a K-dimensional action space. Here, we might loosen the assumptions of the bounds on either action or feature space, to first try to solve easier versions of the problem before adding restrictions on the size of the sets the algorithm can suggest. If we have time after this, we would look into the case where there is correlation between the features.

References

[1] Yanru Guan, Naveen Raman, and Fei Fang. Human decision-making with ai assistance under correlated features. *arXiv preprint arXiv:2606.20628*, 2026.

[2] Gali Noti, Kate Donahue, Jon Kleinberg, and Sigal Oren. Ai-assisted decision making with human learning. In *Proceedings of the 26th ACM Conference on Economics and Computation*, 2025.

Contribution Statement:
Eeshan: came up with problem to look at and formalized it.
Olly: discovered other possible direction which didn't end up as the pursued track, drafted the Background Section
Chirag: helped come up with problem and look into different direction we didn't pursue
Steven: literature search
LLMs were used in evaluating whether this problem was worth solving, and in understanding concepts from literature search.